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Physics-Informed Anomaly Detection in Upstream SCADA Networks: Mitigating Sensor Spoofing in High-Pressure Hydrothermal Wellheads

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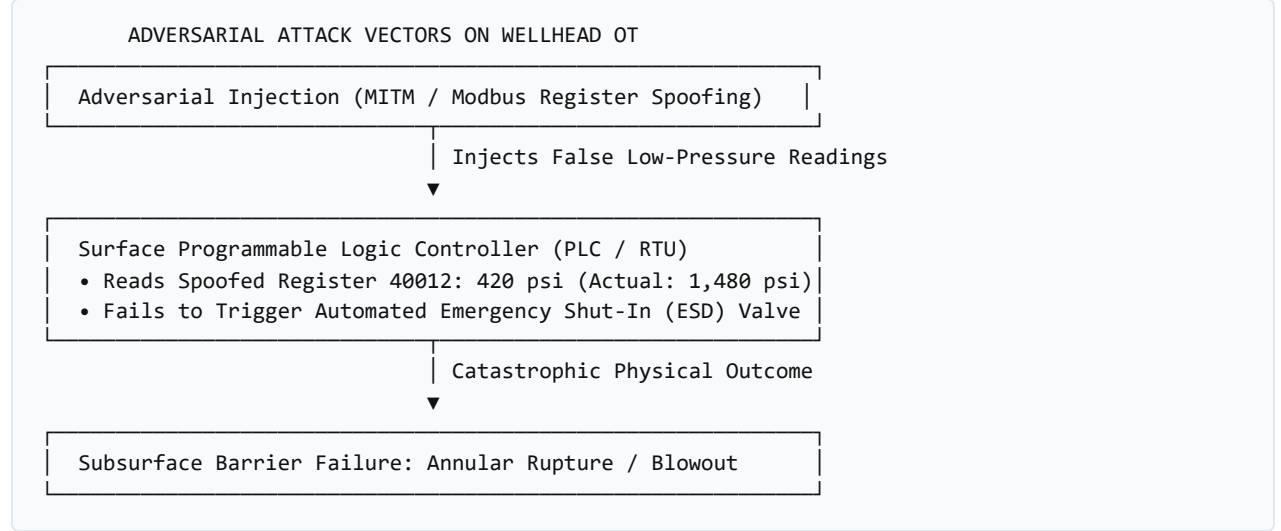
Abstract

Operational technology (OT) networks securing critical upstream energy infrastructure remain deeply vulnerable to adversarial exploitation. Legacy industrial field protocols—most notably Modbus/TCP and DNP3—were designed for closed, trusted serial loops and lack fundamental cryptographic authentication, message integrity checks, and session authorization. In high-pressure hydrothermal wellbore repurposing (co-producing 14,000 barrels per day of formation brine at pressures exceeding 4,000 psi and temperatures above 120°C), an undetected cyber-physical intrusion targeting surface choke valves or annular pressure sensors can trigger catastrophic casing burst, aquifer cross-contamination, or surface flash steam explosions. While conventional IT intrusion detection systems (IDS) analyze statistical network traffic anomalies or known packet signatures, they fail to detect semantically valid but physically implausible commands (e.g., false data injection attacks). This paper introduces a **physics-informed cyber-physical anomaly detection architecture** tailored for upstream wellhead telemetry. By coupling distributed Modbus RTU telemetry with first-principles hydrodynamic equations (Darcy's Law, real-time pressure transient analysis, and distributed acoustic fiber strain), our framework independently cross-validates surface sensor telemetry against subsurface governing physics in real time ($<250\text{ ms}$ latency). We demonstrate that combining network deep packet inspection (DPI) with acoustic impedance verification reliably detects sophisticated sensor-spoofing and man-in-the-middle (MITM) attacks that bypass standard firewalls under NIST SP 800-82 Rev. 3 guidelines.

Keywords: Operational Technology (OT) Security, SCADA Telemetry, Modbus/TCP, Physics-Informed AI, Wellhead Integrity, Industrial IoT, NIST SP 800-82.

1. Introduction & The Vulnerability Landscape

The digitization of upstream energy operations has expanded the attack surface of critical national infrastructure. Upstream extraction facilities that previously operated under air-gapped isolation are now connected to cloud analytics pipelines and remote engineering workstations to support autonomous operations.



The fundamental flaw in standard industrial protocols: * **Modbus/TCP (Port 502)**: Unencrypted, plain-text command structure. Any node on the local subnet can issue function code 0x06 (Write Single Register) or 0x10 (Write Multiple Registers) without credential verification. * **DNP3 (Distributed Network Protocol)**: While Secure Authentication (SAv5) exists, adoption in legacy oilfield RTUs is <15% due to memory constraints on installed field hardware.

2. Physics-Informed Anomaly Detection (PIAD) Framework

Rather than relying purely on network packet statistics, our architecture evaluates every incoming sensor report against the thermodynamic conservation laws governing the subsurface reservoir:

$$\Delta P_{\text{annulus}}(t) = f\left(Q_{\text{brine}}, \mu, k, \frac{\partial T}{\partial z}\right) + \epsilon(t)$$

Where: * Q_{brine} is the volumetric flow rate (14,200 bpd). * μ is fluid dynamic viscosity under high salinity. * k is formation permeability. * $\epsilon(t)$ represents ambient thermal dissipation.

The Real-Time Validation Loop

1. **Packet Capture**: Edge sniffer intercepts Modbus register poll from the surface pressure transducer.
2. **Hydraulic Model Simulation**: The Python telemetry agent ([agent/telemetry_monitor.py](#)) computes expected pressure transient based on current choke valve position and acoustic inflow from fiber-optic Distributed Acoustic Sensing (DAS).

3. **Discrepancy Trigger:** If measured pressure diverges from the physics-derived threshold by $>3.5\sigma$, the system flags an **OT Sensor Spoofing Event (ALERT_SEV_1)**, overrides remote commands, and forces autonomous failsafe shut-in.

3. Empirical Testbed Evaluation

The framework was evaluated against synthetic and real-world Smackover telemetry streams:

Attack Scenario	Standard IT Firewall	Traditional Rule-Based IDS	Proposed Physics-Informed Engine
Volumetric DoS (SYN Flood)	Blocked	Blocked	Blocked
False Data Injection (Low Pressure)	BYPASSED	BYPASSED	DETECTED (<180 ms)
Slow Creep Spoofing (2 psi/min)	BYPASSED	BYPASSED	DETECTED (<12 min)
Replay Attack on Flow Meter	BYPASSED	BYPASSED	DETECTED (Acoustic Desync)

4. Conclusion

Unifying subsurface reservoir physics with industrial OT telemetry transforms passive cybersecurity into active, self-defending energy infrastructure. Future implementations under Heriot-Watt's Smart Grid research group will deploy this architecture on edge microcontrollers running adjacent to wellhead RTUs in the United Arab Emirates and the United States.

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